

## Blow-up and down

We will start by sketching the blow-up in the complex category. Let  $\tau: L \rightarrow \mathbb{CP}^1$  be the tautological complex line bundle

$$L = \{(\ell, p) \in \mathbb{CP}^1 \times \mathbb{C}^2 \mid p \in \ell\}$$

we have the projection  $\pi_2: L \rightarrow \mathbb{C}^2$  and notice that for  $p \in \mathbb{C}^2 \setminus \{0\}$ , then  $\pi_1^{-1}(p)$  is comprised of a single point

Since there is a unique complex line passing through  $p$  and  $0$ .

Hence  $\pi$  defines a diffeomorphism between  $L \setminus \pi_2^{-1}(\{0\})$  and  $\mathbb{C}^2 \setminus \{0\}$ . Notice that  $\pi_2^{-1}(\{0\}) = \mathbb{CP}^1$ .

We have the following result

Prop For a fixed point  $P \in \mathbb{CP}^2$ ,  $\mathbb{CP}^2 \setminus \{P\}$  is diffeomorphic to  $L$ .

Now we are ready for defining the blow-up of a complex surface

Def Let  $S$  be a complex surface with  $P \in S$ , and fix a nbhd of  $P$  in  $S$  biholomorphic to an open subset  $V$  of  $\mathbb{C}^2$  (assume WLOG  $P$  is sent to  $0$ ). Removing  $V$  and replacing it with  $\pi_2^{-1}(V) \subset L$ , we get a new complex manifold  $S'$  (recall that  $U \setminus P$  and  $\pi_2^{-1}(V) \setminus \pi_2^{-1}(0)$  are biholomorphic). The surface  $S'$  is called the blow-up of  $S$  at  $P$ .

Remark Topologically  $S' \cong S \# \mathbb{CP}^2$ , and in general, the blow-up of a 4-manifold is the connected sum with  $\mathbb{CP}^2$ .

In the symplectic world we need to be more careful in following the previous construction.

We study now the special family of Kähler forms  $\tilde{\omega}_\lambda$  on the tautological line bundle  $L$  over  $\mathbb{CP}^1$ , defined

by

$$\tilde{\omega}_\lambda := \pi_2^* \omega_0 + \lambda^2 \tau^* \omega_{FS} \in \Omega^2(L)$$

where  $\omega_{FS}$  is the standard Fubini-Study metric form on  $\mathbb{CP}^1$  (which makes  $\text{vol}(\mathbb{CP}^1) = \pi$ ). Thus  $\tilde{\omega}_\lambda$  is a combination of pullbacks of the standard forms over the projections

$$\begin{array}{ccc} & L & \\ \pi_2 \swarrow & & \searrow \tau \\ (\mathbb{C}^2, \omega_0) & & (\mathbb{CP}^1, \omega_{FS}) \end{array}$$

where, as we already saw, the **blowdown map**  $\pi_2$  is a diffeomorphism over  $\mathbb{C}^2 \setminus \{0\}$  and  $\tau$  is a vector bundle

projection. Moreover, because  $\int_{\mathbb{CP}^1} \omega_{FS} = \pi$ , we find  $\int_E \tilde{\omega}_\lambda = \pi \lambda^2$ , for  $E$  a line in the zero

section of  $L$ .

Denote the closed ball of radius  $\varepsilon$  in  $\mathbb{C}^n$  and its inverse image in  $L$  by

$$B(\varepsilon) := \{z \in \mathbb{C}^n \mid |z| \leq \varepsilon\}$$

$$L(\varepsilon) := \pi_2^{-1}(B(\varepsilon))$$

Thus  $L(\varepsilon)$  is a closed neighbourhood of  $Z = \pi_2^{-1}(0)$ .

Prop  $L(\varepsilon) \setminus Z = \pi_2^{-1}(B(\varepsilon) \setminus 0) = \{(c, v) \in L \mid 0 < |v| \leq \varepsilon\}$  equipped with the symplectic form  $\tilde{\omega}_\lambda$  is symplectomorphic to the spherical shell  $B(\sqrt{\lambda^2 + \varepsilon^2}) \setminus B(\varepsilon)$  in  $(\mathbb{C}^2, \omega_0)$ .

We are now ready for defining the blow-up in the symplectic category. A **symplectic blow-up**  $(\tilde{\Pi}_\lambda, \tilde{\omega}_\lambda)$  of  $(M, \omega)$  of **weight**  $\lambda$  is obtained from a symplectic embedding

$$\psi: B^{2n}(\lambda) \hookrightarrow M$$

Extend  $\psi$  to a symplectic embedding of  $B(\sqrt{\lambda^2 + \varepsilon^2})$  for some  $\varepsilon > 0$  (**Symplectic ball extension Theorem**)

Then replace the image  $\psi(B(\sqrt{\lambda^2 + \varepsilon^2}))$  by the standard neighbourhood  $(L(\varepsilon), \tilde{\omega}_\lambda)$ . Define

$$\tilde{\Pi}_\lambda := (\Pi \setminus \psi(B(\lambda))) \sqcup L(\varepsilon) / \sim$$

where  $([z], z) \in L(\varepsilon) \setminus Z$  is identified with  $\psi\left(\left(\sqrt{1 + \lambda^2 / |z|^2}\right)z\right) \in \Pi \setminus \psi(B(\lambda))$ . One can show that the symplectic form  $\tilde{\omega}_\lambda$  equals  $\omega$  on  $\Pi \setminus \psi(B(\lambda))$  and equals  $\tilde{\omega}_\lambda$  on  $L(\varepsilon)$ . It turns out that the symplectomorphism class of  $(\tilde{\Pi}_\lambda, \tilde{\omega}_\lambda)$  is independent of the choice of the extension used.

## Symplectic blow-down

Suppose that  $X_1, X_2$  are symplectic 4-manifolds and  $F_i \subset X_i$  are 2-dimensional symplectic manifold in them.

Suppose furthermore that  $[F_1]^2 + [F_2]^2 = 0$  and that the genera of  $F_1$  and  $F_2$  are equal. Take an orientation reversing diffeomorphism  $\varphi: F_1 \rightarrow F_2$  and lift it to an orientation-reversing diffeomorphism

$\Psi: \partial \nu F_1 \rightarrow \partial \nu F_2$  between the boundaries of the tubular neighborhoods  $\nu F_i$  (Weinstein symplectic tub. neighborhood theorem). Using  $\Psi$  we can construct  $X_\Psi = (X_1 - \nu F_1) \cup_\Psi (X_2 - \nu F_2)$ . The manifold  $X_\Psi$  is called the (symplectic) normal connected sum of  $X_1$  and  $X_2$  along  $F_1$  and  $F_2$ .

Theorem 10.2.1 Under the above circumstances, the 4-manifold  $X_\Psi$  admits a symplectic structure. Moreover, this structure can be chosen in such a way that the symplectic (Lagrangian) submanifolds of  $X_i - \nu F_i$  are symplectic (Lagrangian) in  $X_\Psi$ .  $\square$

Remark The diffeomorphism type of  $X_\Psi$  depends on the choice of  $\Psi$ .

Theorem 10.2.3 If  $(X, \omega)$  contains a symplectic sphere  $\Sigma$  with square  $-1$ , then  $X$  is symplectomorphic to the blow-up of a symplectic manifold  $Y$ .

Proof Recall that as a  $C^\infty$  4-manifold,  $X$  decomposes as  $Y \# \overline{\mathbb{CP}}^2$  and  $\Sigma$  becomes the exceptional sphere  $\overline{\mathbb{CP}}^1 \subset \overline{\mathbb{CP}}^2$ .

By assumption  $\Sigma$  is symplectic; hence, if we take the symplectic normal connected sum of  $(X, \Sigma)$  with

$(\mathbb{CP}^2, H)$ , (where  $H := \{[z_0:z_1:z_2] \in \mathbb{CP}^2 \mid z_2 = 0\}$ ), i.e. replace  $\nu \Sigma$  with  $\mathbb{CP}^2 - \text{int } \nu H \approx D^4$  and Thm 10.2.1 provides the desired symplectic structure on  $Y$ .  $\square$

We will start today by introducing the concept of Lefschetz fibrations

Def Let  $X$  be compact, connected, oriented, smooth 4-manifold

- a) For  $\partial X = \emptyset$ , a **Lefschetz pencil** on  $X$  is a nonempty finite subset  $B$  of  $X$ , called the **base locus**, together with a smooth map  $\pi: X \setminus B \rightarrow \mathbb{CP}^1$  such that each point  $b \in B$  has an orientation-preserving coordinate chart in which  $\pi$  is given by projectivization  $\mathbb{C}^2 \setminus \{0\} \rightarrow \mathbb{CP}^1$ , and each critical point of  $\pi$  has an orientation-preserving chart on which  $\pi(z_1, z_2) = z_1^2 + z_2^2$  relatively to a suitable smooth chart on  $\mathbb{CP}^1$ . For  $t \in \mathbb{CP}^1$ , the **fiber**  $F_t$  is  $\pi^{-1}(t) \cup B \subset X$

- b) A **Lefschetz fibration** on  $X$  is a map  $\pi: X \rightarrow \Sigma$ , where  $\Sigma$  is a compact, connected, oriented surface and  $\pi^{-1}(\partial \Sigma) = \partial X$ , such that each critical point of  $\pi$  lies in  $\text{int} X$  and has a local coordinate chart as in (a)

Remark Blowing-up the base locus turns a Lefschetz pencil into a Lefschetz fibration over  $S^2$ , with each exceptional sphere a section.

In either case,  $\pi$  has only finitely many critical points, and removing the corresponding singular fibers turns a Lefschetz fibration into a fiber bundle with connected base space. Thus all but finitely many fibers of a Lefschetz fibration (or pencil) are smooth, closed surfaces, all of which have the same diffeomorphism type.

If a generic fiber is connected, its genus will be called the **genus** of the Lefschetz fibration/pencil.

Remark By perturbing  $\pi$  if necessary, one can assume it to be injective on its set of critical points, so that each singular fiber has a unique singularity.

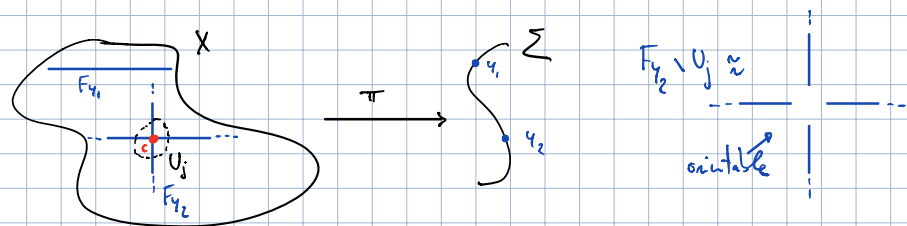


Theorem 10.2.18 Assume that the closed 4-manifold  $X$  admits a Lefschetz fibration  $\pi: X \rightarrow \Sigma$ , and let  $[F]$  denote the homology class of the fiber. Then  $X$  admits a symplectic structure with symplectic fibers iff  $[F] \neq 0$  in  $H_2(X; \mathbb{R})$ . If  $e_1, \dots, e_m$  is a finite set of sections of the Lefschetz fibration, the symplectic form  $\omega$  can be chosen in such a way that all these sections are symplectic.

Proof [sketch] One direction of the above equivalence is easy to prove. If  $X$  carries a symplectic structure  $\omega$  such that a fiber  $F$  is symplectic, then  $\langle [\omega], [F] \rangle > 0$ , in fact  $\omega|_F = \omega_F$  (the symplectic structure of  $F$ ) hence  $\langle [\omega], [F] \rangle = \text{vol}(F) \neq 0$ , hence  $[F] \neq 0$ .

The other direction is more complicated: to construct a symplectic structure on  $X$  assume WLOG that the fibers are connected (cf. Prop 8.1.9) and perturb  $\pi$  so that it becomes injective on the set of critical points.

- let  $X$  as above,  $\exists$  a closed 2-form  $\tilde{\gamma}$  on  $X$  such that if  $F_0$  is a closed surface contained in a fiber (with the induced orientation) then  $\int_{F_0} \tilde{\gamma} > 0$
- let  $\{U_j\}$  be disjoint open balls covering the set of critical points  $C$ . Using the properties of a Lefschetz fibration one can build (compatible) symplectic forms  $\omega_j$  on  $F_y := \pi^{-1}(y)$   $\forall y \in \pi(\bigcup_j U_j)$ . In fact around critical points we can find charts s.t.  $\pi$  is given by  $\pi(z_1, z_2) = z_1^2 + z_2^2$ . Now take  $\omega_j = dx_1 \wedge dy_1 + dx_2 \wedge dy_2$  on  $U_j$ . Since  $F_y \cap U_j$  is a holomorphic curve for  $y \in \pi(U_j)$ , the form  $\omega_j|_{U_j \cap F_y}$  is a symplectic form. Now we need to extend it to the entire fiber. In fact, every fiber away from the critical points is a (possibly non compact) orientable surface, hence once we choose compatible orientation with the one we just constructed we can extend it.



Define  $\omega_y$  for the remaining points  $y \in Z$  in such a way that  $\omega_y$  is symplectic along  $F_y$ . Again, we choose compatible orientations on neighbouring fibres using the fact that  $X, Z$  and the fibres are orientable.

By rescaling away from  $\coprod U_j$ , we can assume that the condition  $[\omega_j] = [\eta_j|_{F_j}]$  holds in  $H_{\text{DR}}^2(F_j) \forall j$ .

- Now one extends these fibrewise forms  $\omega_y$  ( $y \in Z$ ), to closed 2-forms  $\eta_y$  defined on some nbhd of each  $F_y$ , then glue the 2-forms  $\eta_y$  together to get a closed 2-form  $\eta$  on  $X$ . We will skip the details being a general argument involving partition of unity.

Our construction tells us that  $\eta|_{F_j}$  is a convex combination of compatibly oriented area forms on  $F_j$ , hence it's (fibrewise) symplectic.

- Now it's a standard exercise to prove that  $\omega_X := \eta + K\pi^*\omega_Z$  is a symplectic form for  $X$ , for  $K \gg 0$

Addendum: For  $K$  big enough, we can assume that a finite collection of sections will be symplectic, since the pull-back  $\pi^*\omega_Z$  is symplectic along sections, and for  $K \gg 0$  it dominates in  $\omega_X$ .

Thanks to this addendum, the statement of the Theorem can be extended to Lefschetz pencils as well

Corollary: If a 4-fold  $X$  admits a Lefschetz pencil, then it has a symplectic structure.

Proof: By blowing up  $X$  in the  $n$  points of the Base locus we get a Lefschetz fibration  $X \#_n \overline{\mathbb{CP}}^2 \rightarrow \mathbb{CP}^1$

Since  $B \neq \emptyset$ , there are sections (the exceptional spheres) intersecting each fiber non-trivially.

Hence the homology class of the fiber is non-trivial, so by the Theorem, the blown-up mld  $X \#_n \overline{\mathbb{CP}}^2$  admits a symplectic structure for which the exceptional spheres are symplectic.

Now symplectically blowing down the exceptional spheres results in a symplectic structure on the mld  $X$ . □